DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer building a provider-facing **SMART on FHIR** clinical decision support app (Next.js + Firebase/GCP) from the ground up, surfacing covered, guideline-backed medication alternatives in the EHR prescribing flow to prevent prior authorization.
- Extending a **Firebase Cloud Functions** backend that assembles alternatives from **RxNorm** normalization, formulary/prior-auth logic, and PubMed/guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility and coverage.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's text-analysis platform (**Python**, **FastAPI**, sentence-transformers, React) as a self-serve research tool, owning architecture, the production design doc, and all technical decisions end to end.
- Built a YAML model registry as a single source of model truth and a versioned construct library (**99 constructs from 38 questionnaires**), plus structured, machine-readable data-quality checks; grew the backend to **40 hermetic tests** running in CI without ML dependencies.
- Built per-run reproducibility exports in **Python**, an offline-runnable script generated purely from stored run metadata (item wordings, model revision, package pins), so any analysis can be re-run without platform access.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **$340K+** in revenue.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Owned backend security and reliability for the eTriage stack, implementing **AWS KMS** encryption for clinical PII, hardening authentication and rate limiting, adding integration tests, and remediating **20+ VAPT findings** to keep the multi-tenant service safe to change.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built an object-oriented **Python/FastAPI** service that turns open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**, instead of a single opaque call; hardened with **150+ automated tests** and provider routing/fallback across LLM backends.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Optimized a **Spark/PySpark** ETL pipeline over a **~1.4B-row** dataset, cutting end-to-end runtime **~25%** (17–19 min to 14.3 min) via broadcast joins (shuffle minutes to ~0.05s), Adaptive Query Execution, skew-join handling, and schema-aware column pruning, using Spark UI stage profiling to localize bottlenecks.

Cache- and SIMD-Aware Matrix Multiplication (GitHub) C++, AVX/SIMD, OpenMP, Linux perf

- Optimized C++ matrix multiplication from a naive baseline through cache-aware tiling, manual **AVX/SIMD** vectorization via `__m256` intrinsics, register-aware micro-kernels, and **OpenMP** multithreading, profiling L1/L3/TLB misses with `perf` and inspecting generated assembly.

Publications

- *"Alzheimer's Disease Classification using Transfer Learning,"* IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, C++, TypeScript, JavaScript, Java, SQL, Bash

Backend & APIs: FastAPI, Node.js, Express, REST APIs, async services & queues, pytest, integration testing

Data & Systems: PySpark, Spark UI, distributed data processing, Apache Kafka, Hadoop/MapReduce, PostgreSQL, MongoDB

Performance: Linux perf, AVX/SIMD, OpenMP, cache-aware tiling, tail-latency analysis, SQLite, Redis

Cloud & DevOps: AWS, GCP, Azure, Docker, Git, GitHub Actions, Kubernetes, CI/CD